Yaroslav V. Baienko
Attorney-at-Law, Legal Engineer & Applied Cryptography Researcher
ORCID: 0009-0006-7498-2133

Dear Editor-in-Chief,

Please consider the manuscript **Fragile Authentication of PDF Signatures via Bit-Exact Quantization Index Modulation on Vector Geometry** for publication in **IEEE Transactions on Information Forensics and Security** as a regular paper.

Problem. A PDF signature verifies the bytes it covers, not the page a human sees; the visible seal is typically not cryptographically bound to the signed content, and published attacks on PDF signatures exploit this gap. The paper treats the problem as fragile watermarking: can the visible mark itself carry a bit-exact digest, so that the mark a person sees is also the mark that is verified?

Contribution. The digest is embedded, bit-exactly, into the geometry of a visible seal (nine triangles, 27 scalar degrees of freedom) using keyed dithered QIM over an integer coordinate lattice with an exact IEEE-754 argument; a disjoint carrier degree of freedom carries a per-recipient identifier. No external QR/barcode symbol is added. The paper reports the implementation precondition that matters in practice — the PDF writer must preserve at least nine decimal places and the verifier must parse coordinates as text — and shows that common workflows (five-decimal writers, 32-bit-float drawing APIs) destroy the lattice.

Evaluation. A 180-file corpus from three generators (LibreOffice, Typst, Chromium print) in native, flattened and print-scan-simulated variants, plus ten public project documents and ten independent public documents (five NIST standards, five public IRS forms); 1,800 + 200 stamped PDFs with stream-level attacks; carrier ablation (incidence geometry vs generic polyline vs grid); and same-carrier baselines re-implemented at both slot and PDF level. Capacity-limited baselines are reported as such; the paper does not claim cryptographic security from a watermarking argument and explicitly complements, rather than replaces, PAdES/eIDAS signatures.

Preprints and prior venues. All posted preprints of this research line are disclosed in the submission: Zenodo concept DOI 10.5281/zenodo.23429017 (latest version 10.5281/zenodo.23560975, v2.1.0-rev1), SSRN DOI 10.2139/ssrn.7631458, and Figshare DOI 10.6084/m9.figshare.34697025. The submitted manuscript is a substantially reframed and expanded v3; no posted preprint is identical to it, and the earlier editions are cited as related prior work (Baienko, Zenodo 2026, DOI 10.5281/zenodo.23560975) with the relationship explained. A Ukrainian patent application (a 2026 06266) covers the sealing method. The manuscript is not under consideration elsewhere. The evaluation harness, corpus manifest and seeds are available for reproducibility.

Thank you for your consideration.

Sincerely,
Yaroslav V. Baienko